



Supporting Information for

Distinct latitudinal patterns of molecular rates across vertebrates

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Material and Methods

Sequence data and alignments

A comprehensive search of complete and near-complete mitochondrial genomes in GenBank was conducted using the search terms “mitochondrion genome” in the “All Field” category and “Mammalia” OR “Aves” OR “Amphibia” OR “Reptilia” OR “Actinopterygii” in the “Organism” category up to October 2021. Duplicate sequences were removed, and the 12 protein-coding genes on the heavy strand were aligned by codon. A maximum likelihood (ML) tree was constructed using concatenated alignments in RAxML 8.2.4 (1) with 100 rapid bootstrap searches. Spurious sequences identified during the preliminary analysis were removed, and one genome per species was selected. For species with multiple genomes, the longest, highest-quality genome with the fewest missing or ambiguous codons was selected; if genomes were of equal quality, one was chosen at random. Sequences were realigned by codon for each gene using MUSCLE 3.8.31 (2), and codons with alignment gaps in any species were manually deleted in MEGA X (3). After quality control, the final dataset included sequences from 1113 birds, 1354 mammals, 393 reptiles, 225 amphibians and 2539 ray-finned fishes, with alignment lengths ranging from 10,041 to 10,656 base pairs.

Nuclear coding regions of birds and mammals annotated using TOGA (4) were downloaded, and one-to-one orthologs were extracted. For ectotherms with limited genomic data, ultraconserved elements (UCEs) of fishes (5), amphibians (6), and reptiles (7) were downloaded, and coding regions were extracted using BLAST. Orthologs were aligned with MUSCLE 3.8.31 (2). To identify potential misalignments or misidentified orthologs, we filtered the dataset based on maximum likelihood (ML) trees, excluding genes with branch lengths exceeding 10% of the total tree length (i.e., the sum of all branch lengths). This relative branch length test helps detect anomalously long branches indicative of alignment errors. Ambiguous regions were further identified and removed using Gblocks (8) based on codon structure. The final dataset included sequences from 468 birds, 421 mammals, 279 reptiles, 272 amphibians, and 262 ray-finned fishes, with alignment lengths ranging from 0.25 to 3.24 Mb.

Distribution, latitude, ambient temperature and life-history traits

We obtained present-day distribution maps from the BirdLife International Database (9), IUCN Red List of Threatened Species (10), AquaMaps (11) and RepFocus (12). For species lacked expert-checked distribution maps, occurrence points were collected from the Global Biodiversity Information Facility (GBIF, <https://www.gbif.org>), VertNet (<http://vertnet.org>), FishNet2 (<http://www.fishnet2.net>), Ocean Biogeographic Information System (OBIS), museum records and literatures. We then used Maxent (13) to predict their distribution maps using climatic and environmental layers.

Latitude and ambient temperature were estimated across the full distribution range. Latitude was calculated as the centroid of distribution maps, while ambient temperature was the average temperature across the full range. For terrestrial and freshwater species, mean annual temperature (BIO1) was derived by overlaying geographic maps with raster climatic layers at 2.5 arc-minute resolutions from the CHELSA global climate data V2.1 (14). For marine species, mean surface temperatures were obtained from Bio-ORACLE v2.0 marine data layers (15) based on geographic maps. Annual mean temperatures of marine animals were roughly estimated based on the depth of their habitat and surface water temperature. For depths < 200 m, temperature was assumed to be the same as surface water temperature. For depths between 200 and 1000 m, temperature was estimated using the formula $T_D = \frac{800(T_s - 4)(1000 - D)}{800} + 4$ (where T_s is the surface water temperature and D is the depth of distribution). For depths > 1000 m, temperature

remains constant at 4°C. The influence of hydrothermal vents on deep-sea temperatures was not considered.

We collected data on life-history traits, including body mass, maximum longevity, generation time, fecundity and basal metabolic rate (BMR), to examine their effects on molecular rates (*SI Appendix*, Table S27–S28). Generation time was defined as the age at first breeding for endotherms and the age at sexual maturity for ectotherms. Fecundity was measured as clutch/litter size $\times$ clutches/litters per year for mammals and birds, and as the number of eggs per year for fishes, amphibians and reptiles. For birds, we collected body mass and migration status from AVONET (16), longevity and age at first breeding from Bird et al. (17) and clutch size from Jetz et al. (18). For mammals, body mass, longevity and age at first breeding were sourced from Pacifici et al. (19), with litter size from the literature (20–22). For amphibians, body mass, length, longevity, age at sexual maturity and clutch size are sourced from Olveria et al. (23). For reptiles, body mass and snout-vent length (SVL) data were obtained from Feldman et al. (24), longevity from Scharf et al. (25), and the age of sexual maturity and clutch size data from Meiri (26). Life-history traits of fishes were collected from FishBase (27). Missing data are supplemented from the “Handbook of the Birds of the World” (28), AnAge (<https://genomics.senescence.info/species/index.html>), WorldSpecies (<https://worldspecies.org>), Animal Diversity Web (ADW, <https://animaldiversity.org>) and other sources (*SI Appendix*, Table S27–S28). For ectotherms, body mass was estimated using body length or snout-vent length (SVL) using family-specific log-transformed relationships. Mass-specific BMR, body mass and body temperature of birds were obtained from Fristoe et al. (29).

Methods used to account for non-independence in datasets

Our analyses accounted for three sources of non-independence in the phylogenetic generalized linear mixed models (PGLMMs).

1. Phylogenetic non-independence due to shared ancestry

Molecular rates exhibit strong phylogenetic signals due to shared evolutionary history. To account for this, we incorporated phylogenetic relatedness as a random effect by including a phylogenetic covariance matrix derived from the divergence time tree using the `inverseA()` function in the R package `MCMCglmm` v2.34 (30). For the best-fit trait evolutionary model, the phylogenetic covariance matrix was converted from a rescaled time tree, with all branch lengths transformed according to the best-fit parameters determined by the `fitContinuous()` function.

2. Non-independence within similar habitats

Species inhabiting similar environments may experience correlated evolutionary trajectories due to shared ecological and environmental conditions. To account for this, we initially included habitat as a random effect, with both random intercept and slope. However, incorporating random slopes substantially increased AIC values across models. Consequently, we retained only random intercepts for habitat in the final models.

3. Non-independence arising within lineages used for estimating molecular rates

Although molecular rates were estimated from phylogenetically independent lineages, residual non-independence may persist within lineages. To address this, we included lineages as a random effect in the PGLMMs.

Identification of the best-fit models for molecular rates

By default, PGLMM assumes a Brownian motion (BM) (31) model of trait evolution. To identify the best-fit model for variation in dN and dS, we compared the BM model with two alternative models: the Ornstein-Uhlenbeck (OU) model (32) and the Early-burst (EB) model (33), also known as the

accelerating-decelerating (ACDC) model(34). The OU model describes a random walk with a central tendency and an attraction strength proportional to the parameter alpha (α). The EB model allows evolutionary rates to change exponentially through time with the parameter a (acceleration: $a > 1$; BM: $a = 1$; deceleration: $a < 1$). We fitted all three different models using maximum likelihood estimation (ML) using the `fitContinuous()` function in GEIGER (35), and selected the best-fit model based on Akaike Information Criteria (AIC). The time tree was then transformed according to the best-fit model and its optimal parameters using the `rescale()` function in the APE (36) to transform the time tree according to the best-fit model and its optimal parameters. The transformed tree was utilized to adjust phylogenetic covariance matrix, which was incorporated into the PGLMMs.

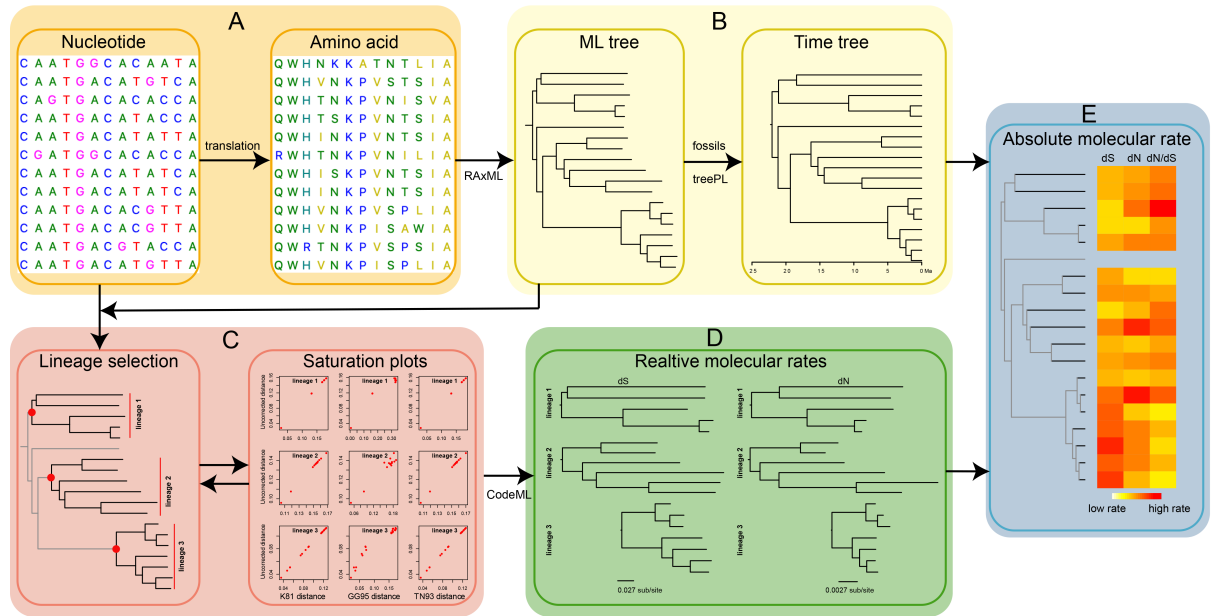


Fig. S1. Three-step phylogeny-based approach to estimate absolute molecular rates. (A) Nucleotide sequences were translated into amino acid sequences, which were then used for phylogenetic inference with RAXML. (B) Divergence time estimation was performed based on the maximum likelihood (ML) amino acid tree and fossil calibration points using treePL. (C) Subclades were identified from orders to lower taxonomic levels along the ML tree, based on the combined results of the K81, GG95, and TN93 models, until their saturation plots indicated moderate saturation at 0.4 substitutions/site. (D) Codeml was used to estimate relative synonymous (dS) and nonsynonymous (dN) substitution rates for each selected subclade. (E) Absolute species-specific molecular rates and dN/dS ratios were calculated based on terminal branch relative molecular rates and divergence times.

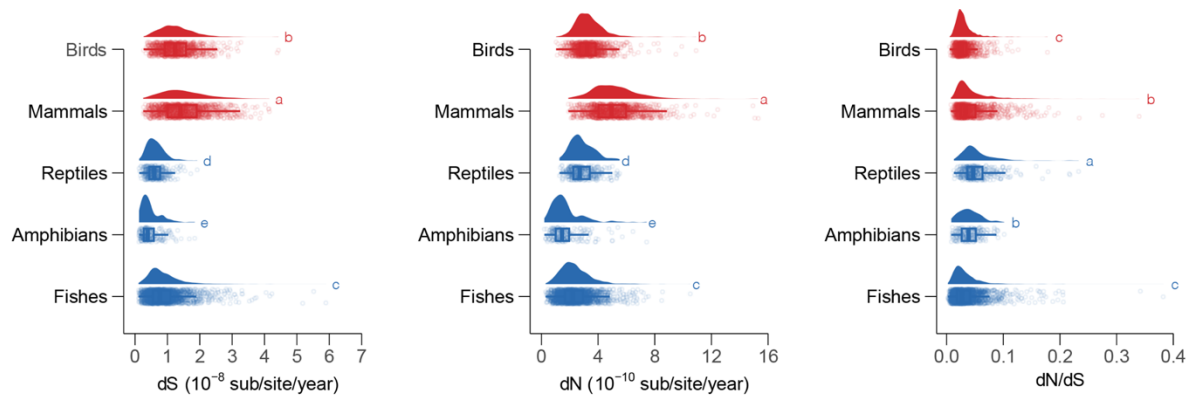


Fig. S2. The distribution of synonymous (dS), nonsynonymous (dN) substitution rates and dN/dS for mitochondrial genes in vertebrates. Letters indicate statistical significance from the Kruskal–Wallis test with Bonferroni-correction P values ($\alpha = 0.05$).

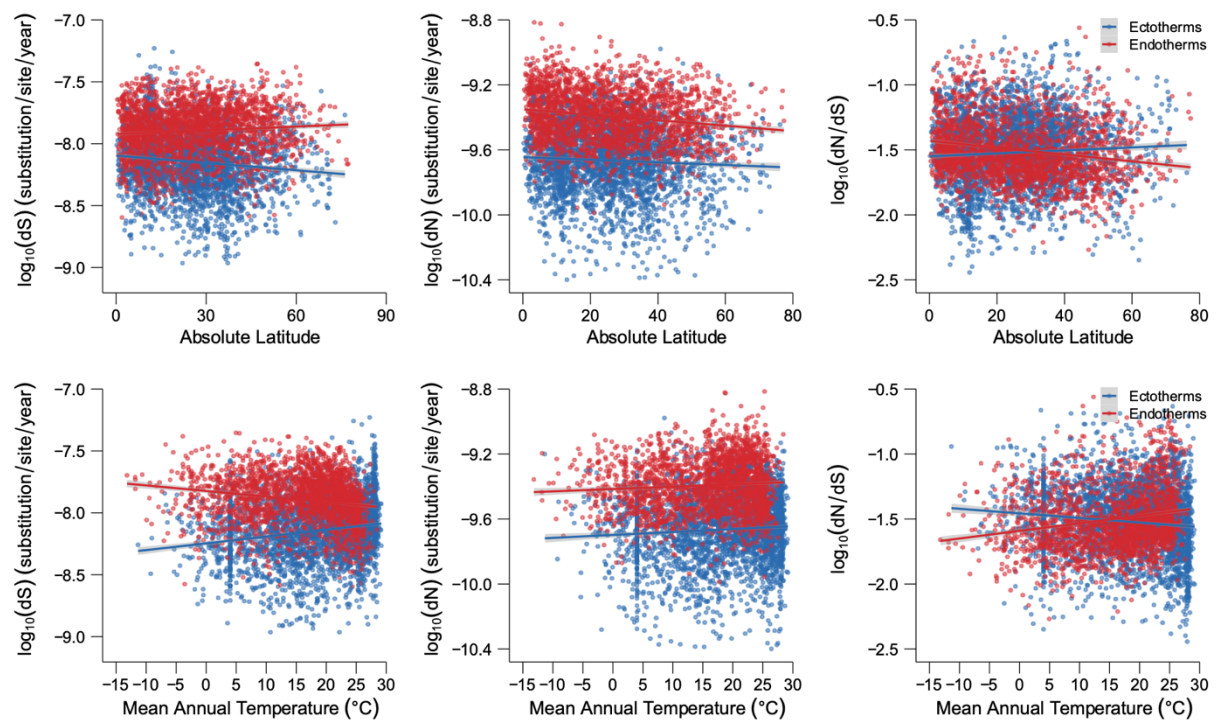


Fig. S3. Relationships between species-specific dS, dN, dN/dS and latitude/temperature for mitochondrial genes in endotherms and ectotherms. Trend lines are based on ordinary least-squares regression for illustrative purposes.

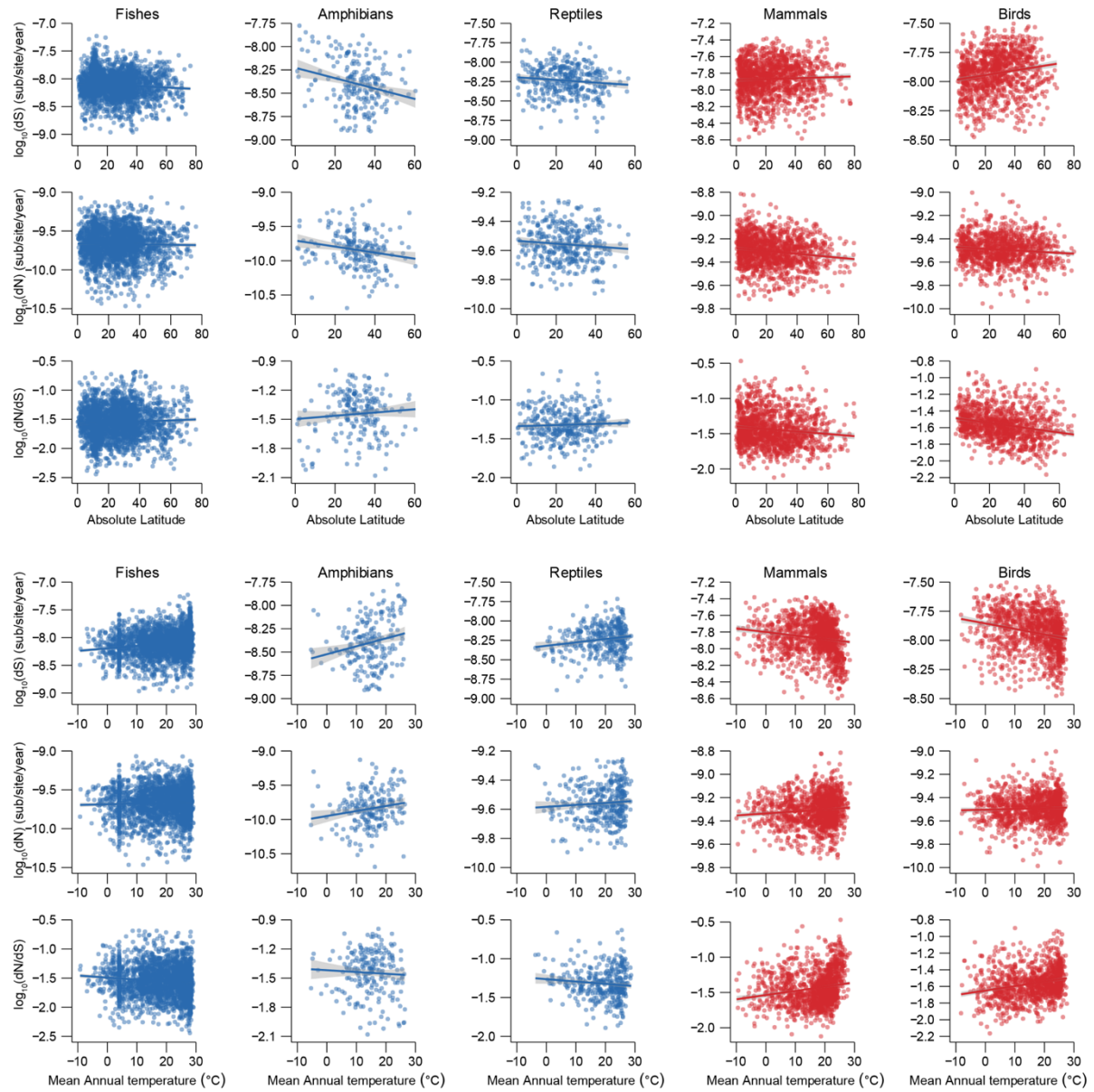


Fig. S4. Relationships between species-specific dS, dN, dN/dS and latitude/temperature for mitochondrial genes by class. Trend lines are based on ordinary least-squares regression for illustrative purposes.

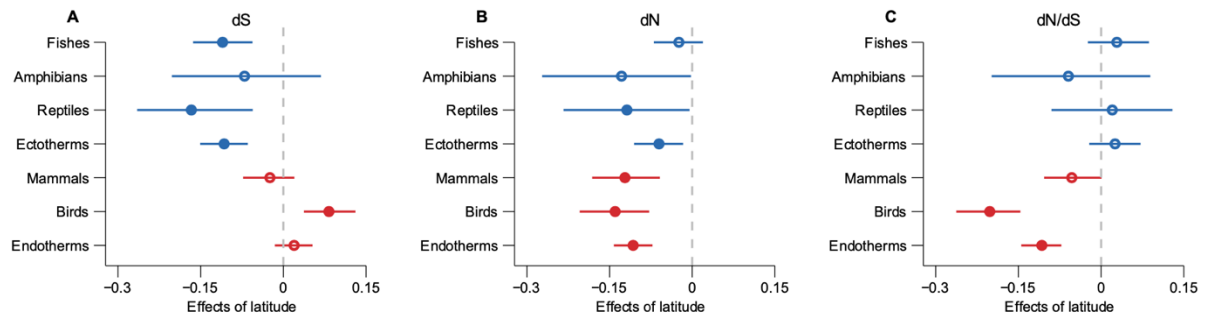


Fig. S5. Effects of absolute latitude on dS, dN and dN/dS for mitochondrial genes. Bars represent 95% confidence intervals, with filled points indicating statistical significance ($P_{\text{MCMC}} < 0.05$).

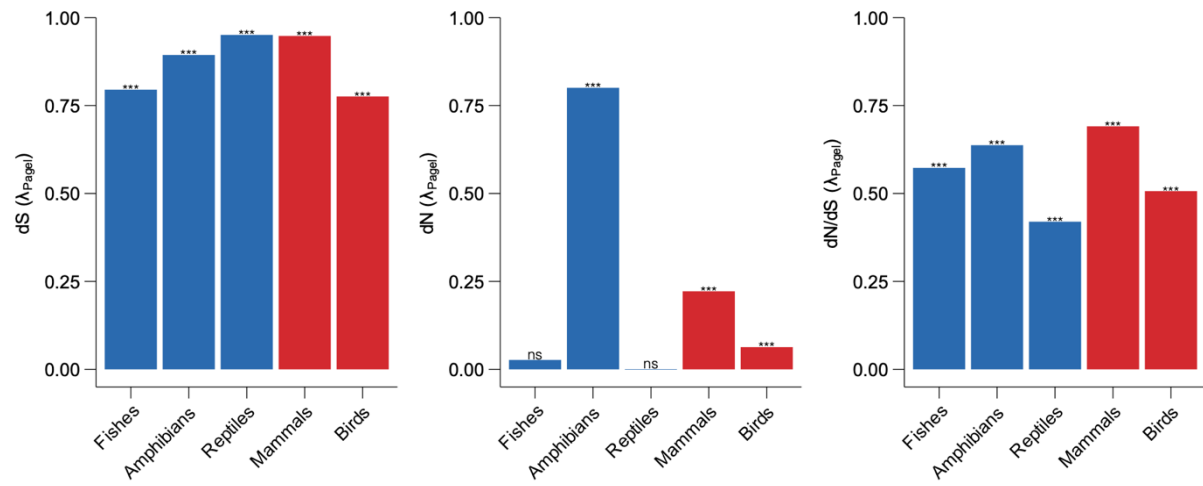


Fig. S6. Phylogenetic signals of dS, dN and dN/dS for mitochondrial genes in each class of vertebrates. The phylogenetic signals were estimated by λ_{Pagel} .

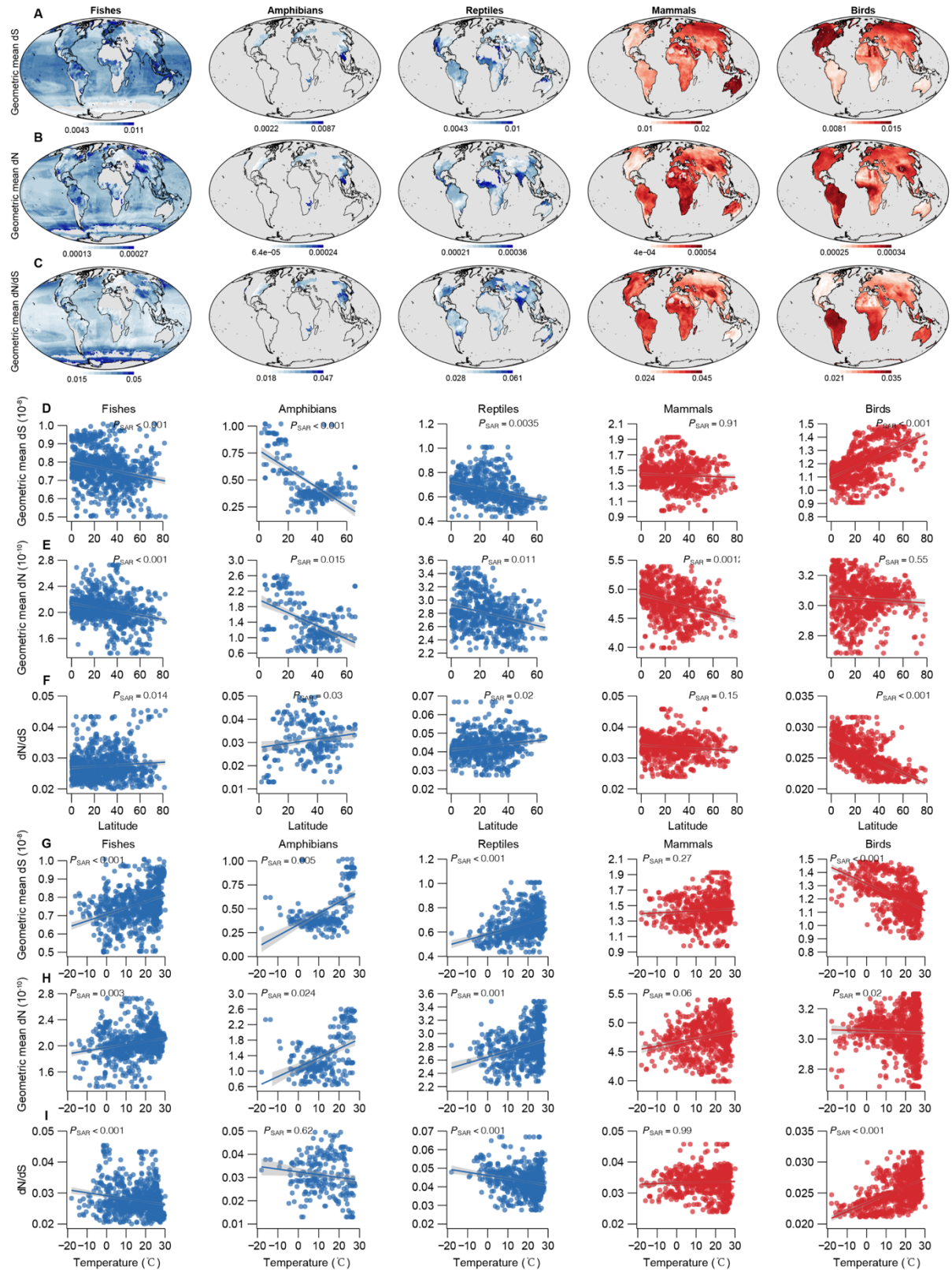


Fig. S7. Latitudinal and temperature dependence patterns of the geometric means of mitochondrial molecular rates or dN/dS in assemblages by class. Trend lines are indicative, based on ordinary least-squares regression. Statistical significance is indicated by spatial simultaneous autoregressive (SAR) models to account for spatial autocorrelations.

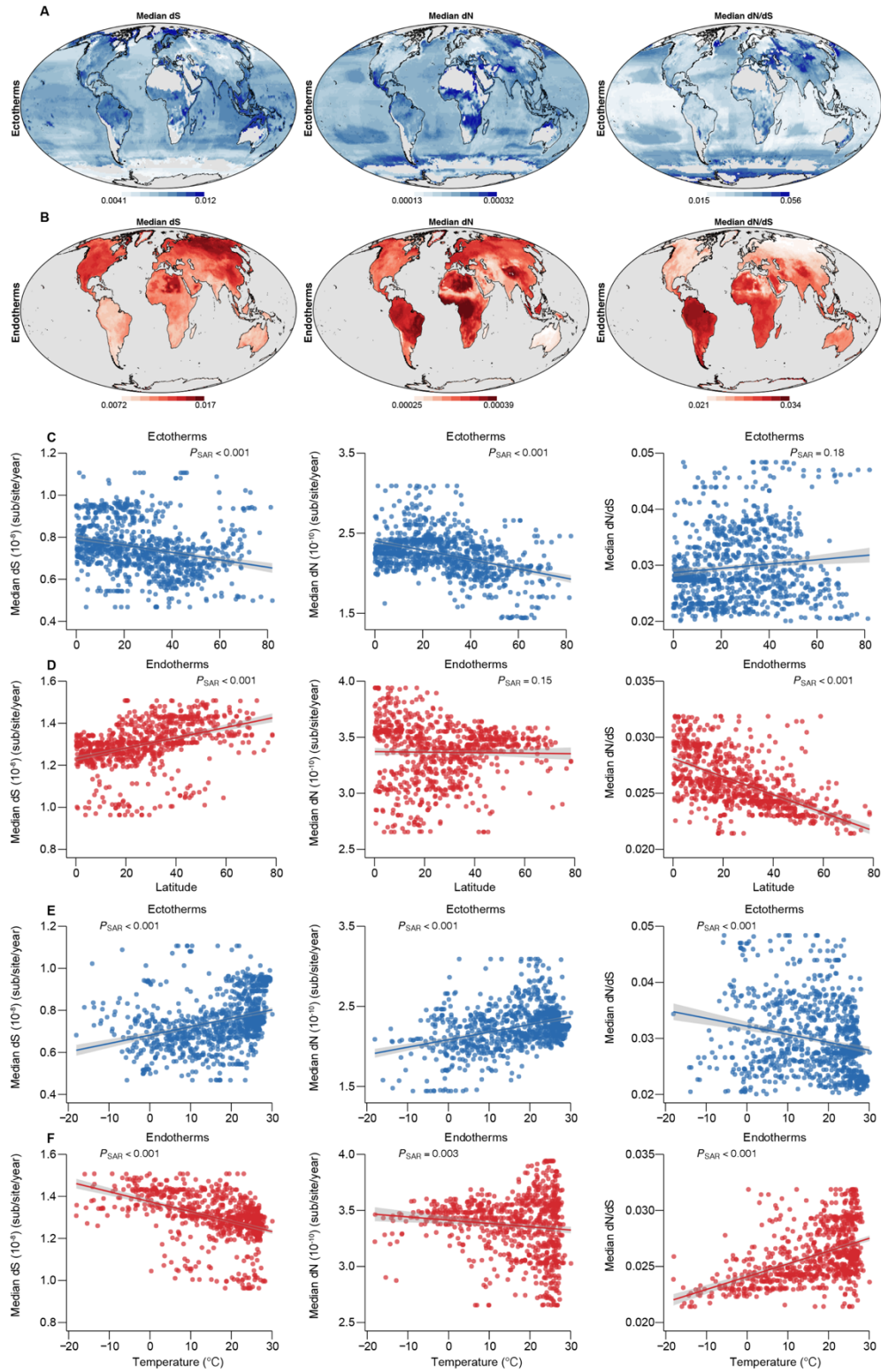


Fig. S8. Latitudinal and temperature dependence patterns of median dS, dN and dN/dS for mitochondrial genes in endotherm and ectotherm assemblages. (A, B) Global distribution of median dS, dN, and dN/dS for endotherms and ectotherms, with grids <5 species excluded. (C, D) Relationships between median dS, dN, dN/dS and absolute latitude in ecoregions. (E, F) Relationships between median dS, dN, dN/dS and temperature in ecoregions. Trend lines are indicative, based on ordinary least-squares regression. Statistical significance is indicated by spatial simultaneous autoregressive (SAR) models to account for spatial autocorrelations.

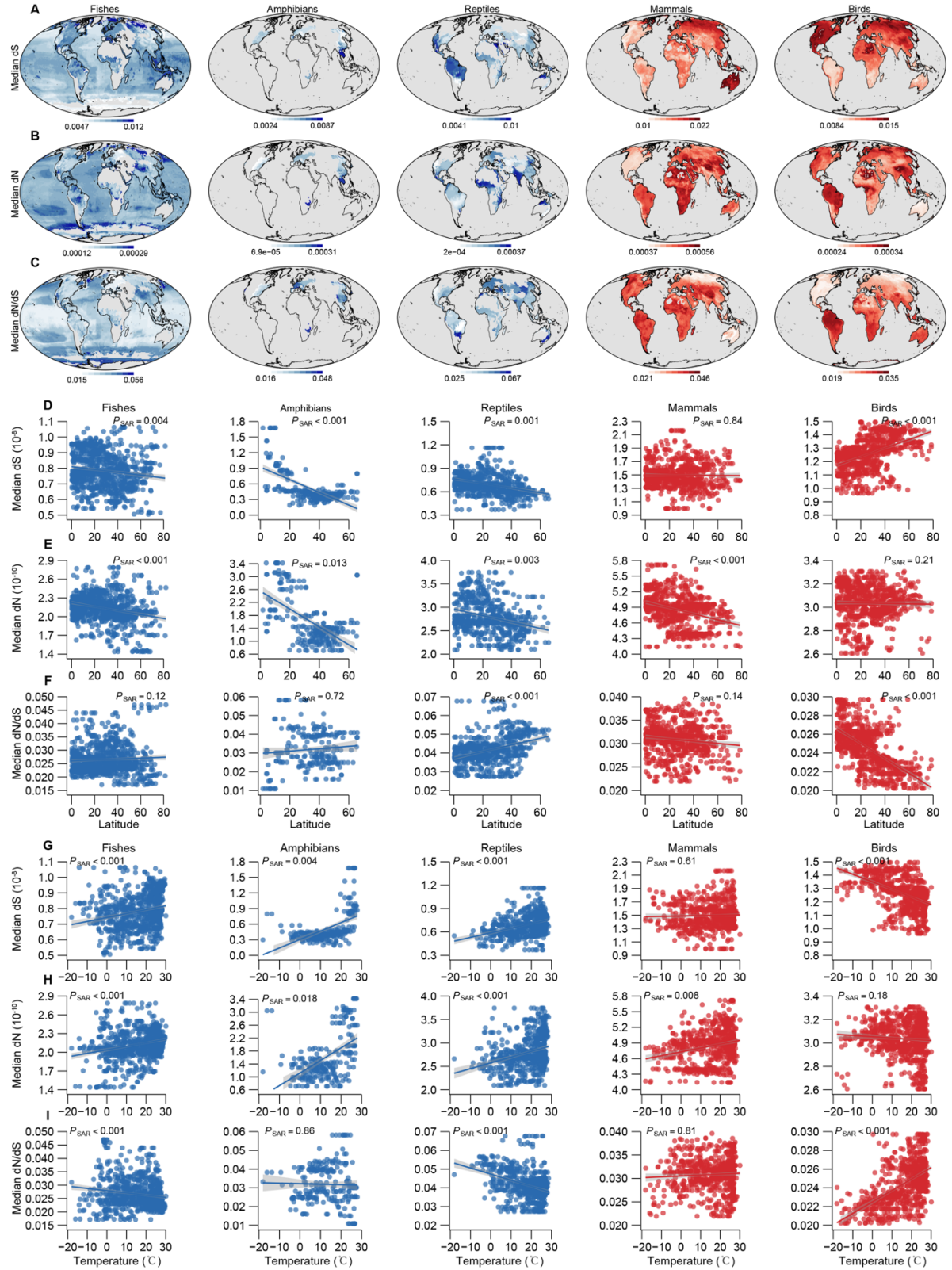


Fig. S9. Latitudinal and temperature dependence patterns of median dS, dN and dN/dS for mitochondrial genes in assemblages by class. Trend lines are indicative, based on ordinary least-squares regression. Statistical significance is indicated by spatial simultaneous autoregressive (SAR) models to account for spatial autocorrelations.

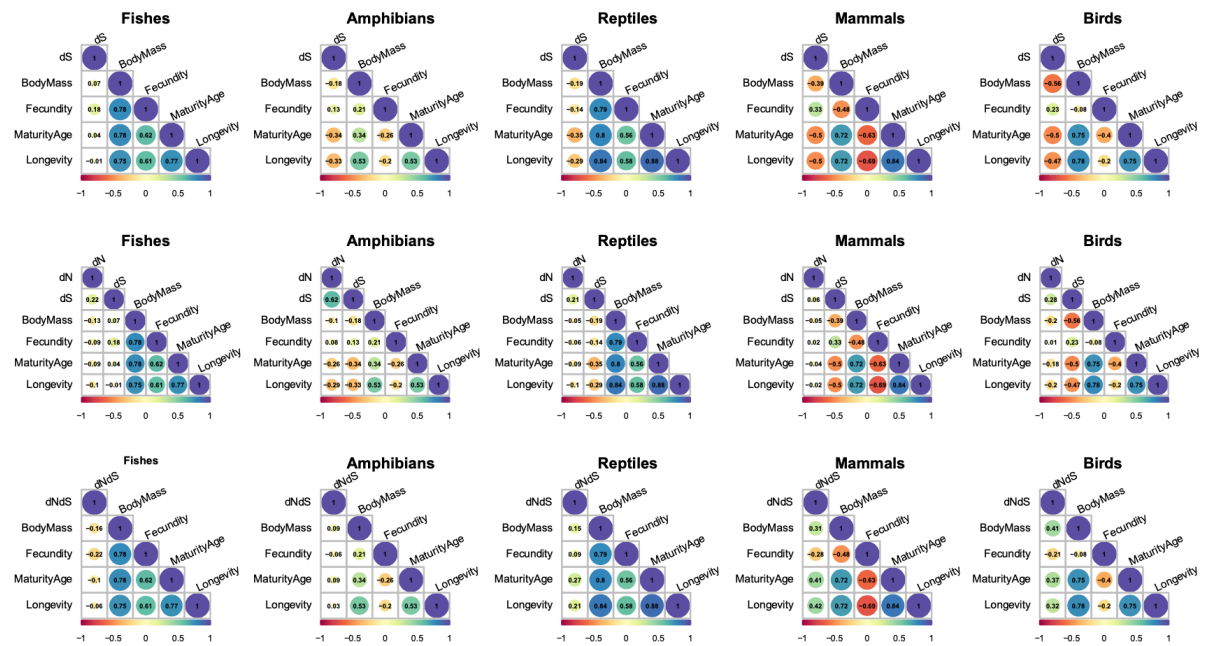


Fig. S10. Correlation plots showing strong associations between dS, dN, dN/dS and life-history traits. Coefficients are estimated using Pearson's correlation of log-transformed dS, dN, dN/dS and life-history traits.

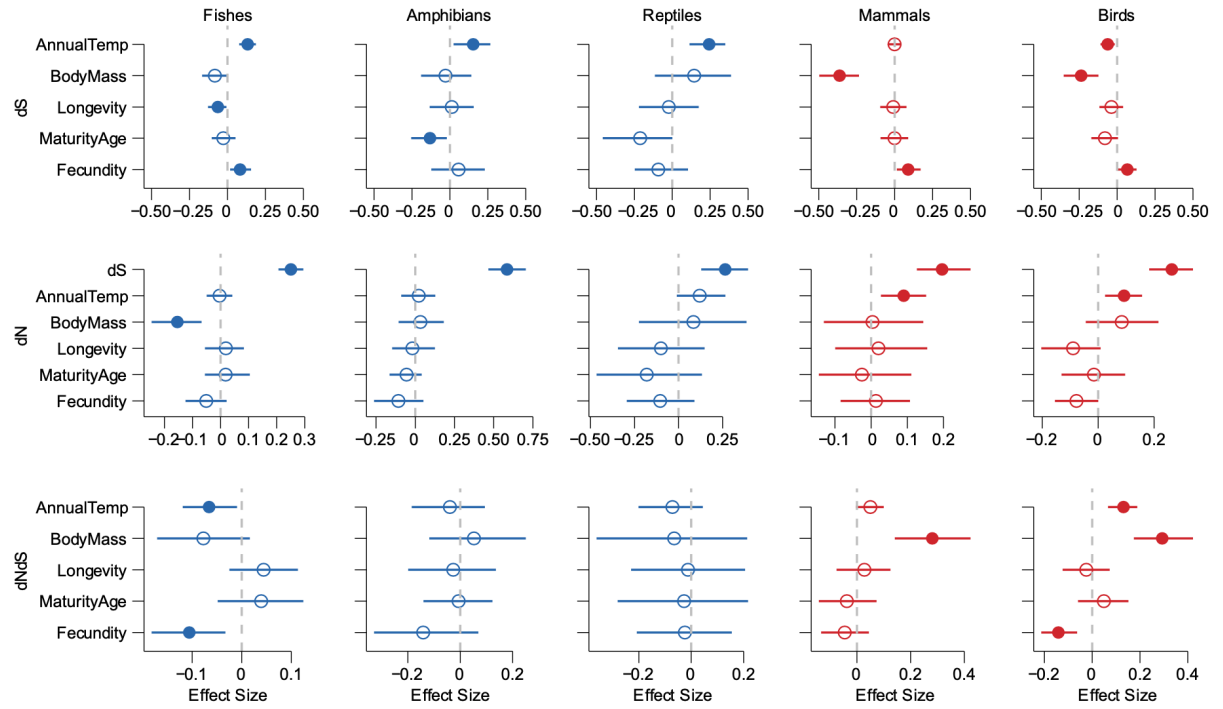


Fig. S11. Impacts of ambient temperature on dS, dN and dN/dS for mitochondrial genes after accounting for life-history traits and phylogenetic signals. Effect sizes of fixed factors with 95% confidence intervals were using standardized slopes from PGLMMs. Positive values indicate a positive association with molecular evolutionary rates, while negative values indicate a negative association. Statistically significant differences ($P_{MCMC} < 0.05$) are indicated by filled points.

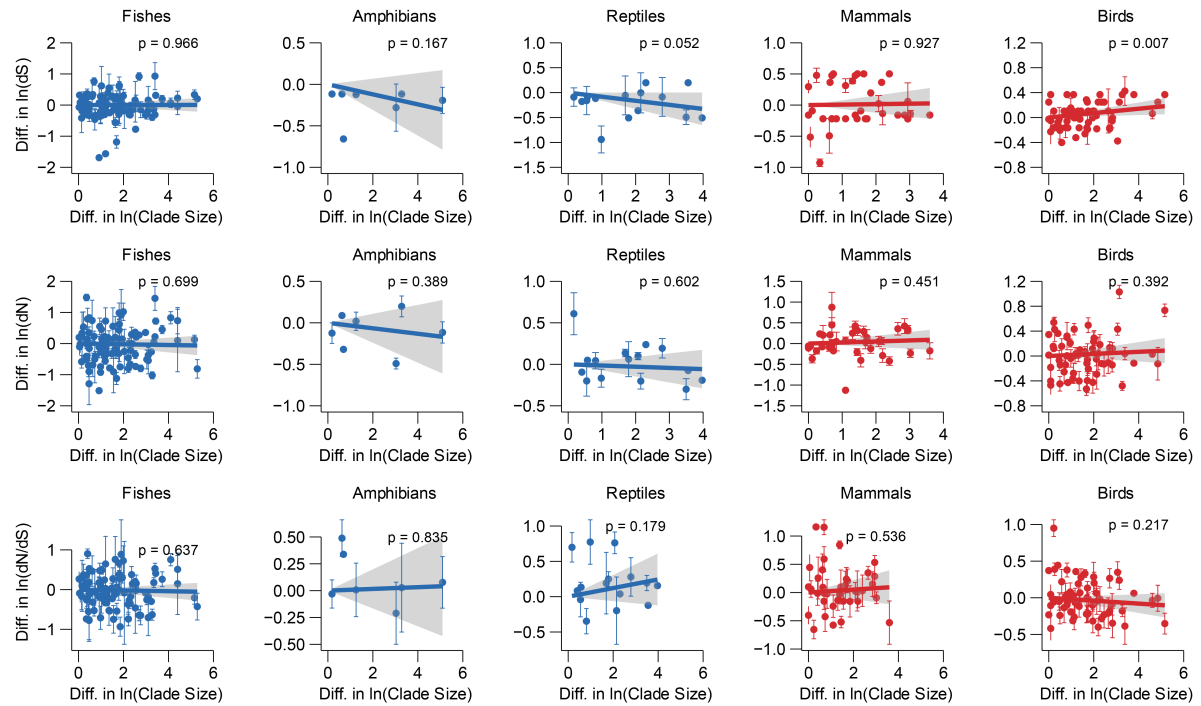


Fig. S12. Correlations between dS, dN, dN/dS and diversification rates for mitochondrial genes in vertebrates. Relationships are shown by differences in molecular rates and differences in clade size between two lineages in each sister pair. Error bars show standard errors in each sister lineage. Statistically significant differences are indicated by ordinary least squares regressions through the origin.

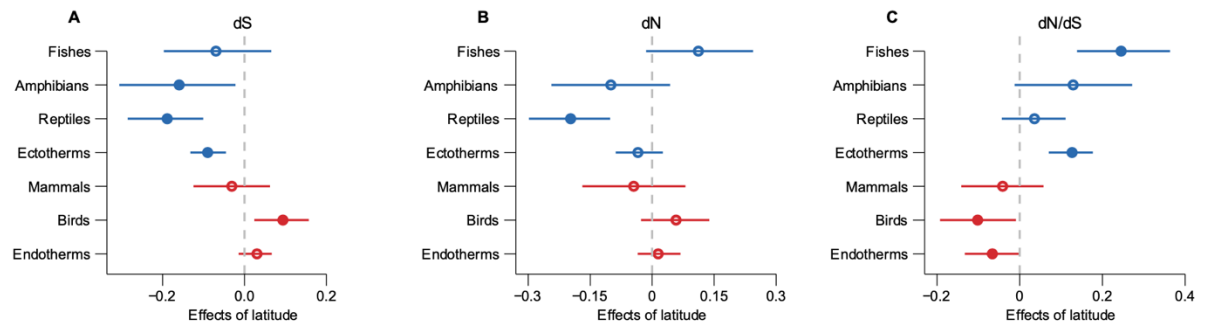


Fig. S13. Effects of absolute latitude on dS, dN and dN/dS for nuclear genes. Bars show 95% confidence intervals. Statistical significance ($P_{\text{MCMC}} < 0.05$) is indicated by filled points.

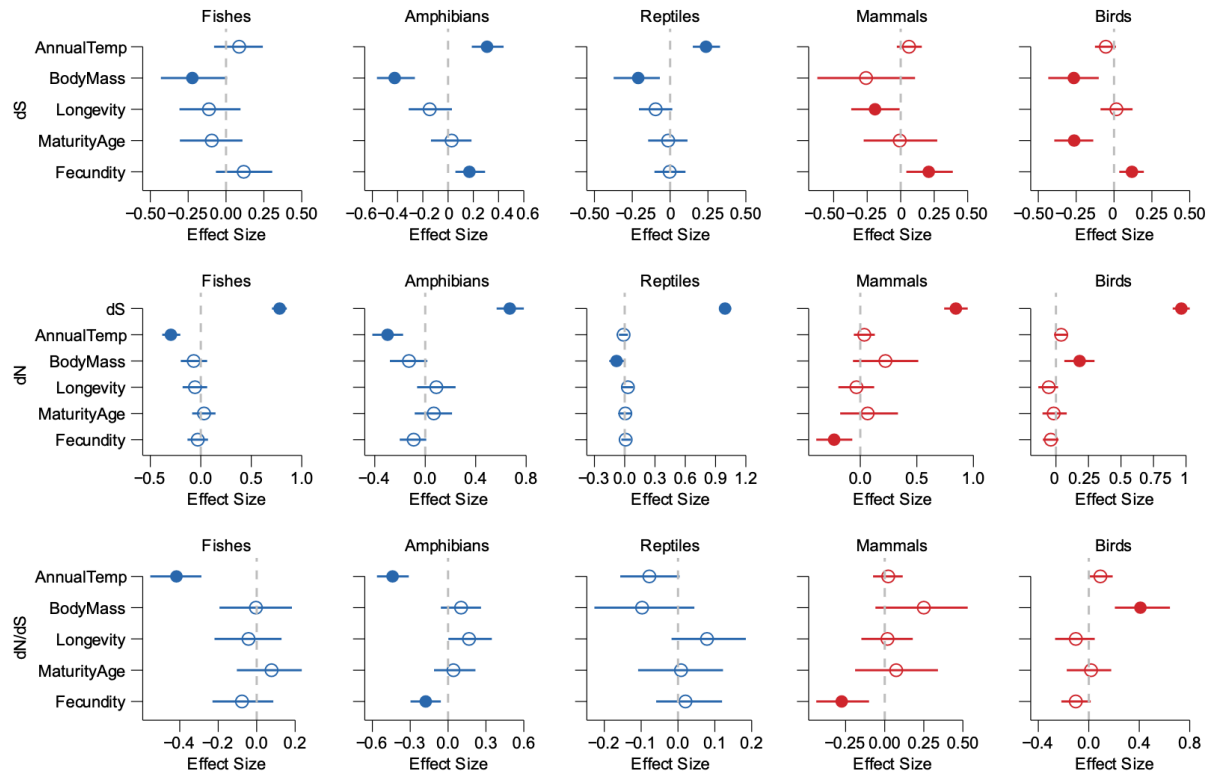


Fig. S14. Impacts of ambient temperature on dS, dN and dN/dS for nuclear genes after accounting for life-history traits and phylogenetic signals. Effect sizes of fixed factors and 95% confidence intervals are estimated using standardized slopes from PGLMMs. Positive values indicate a positive association with molecular evolutionary rates, while negative values indicate a negative association. Statistically significant differences ($P_{\text{MCMC}} < 0.05$) are indicated by filled points.

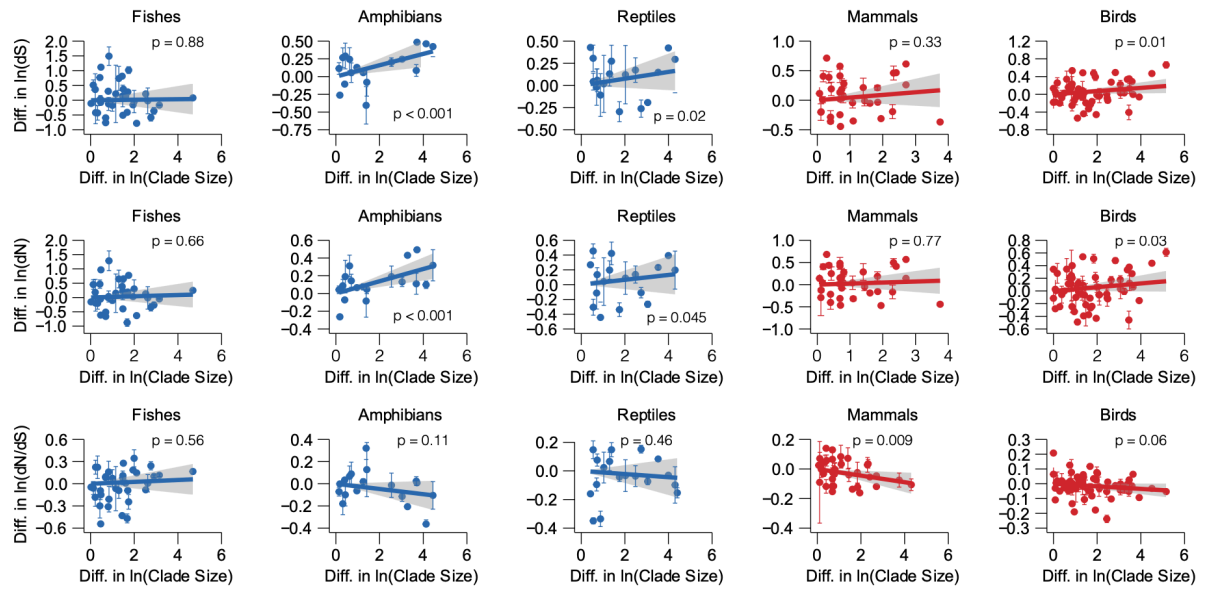


Fig. S15. Correlations between dS, dN, dN/dS and diversification rates for nuclear genes in vertebrates. Relationships are shown by differences in molecular rates and differences in clade size between two lineages in each sister pair. Error bars show standard errors in each sister lineage. Statistically significant differences are indicated by ordinary least squares regressions through the origin.

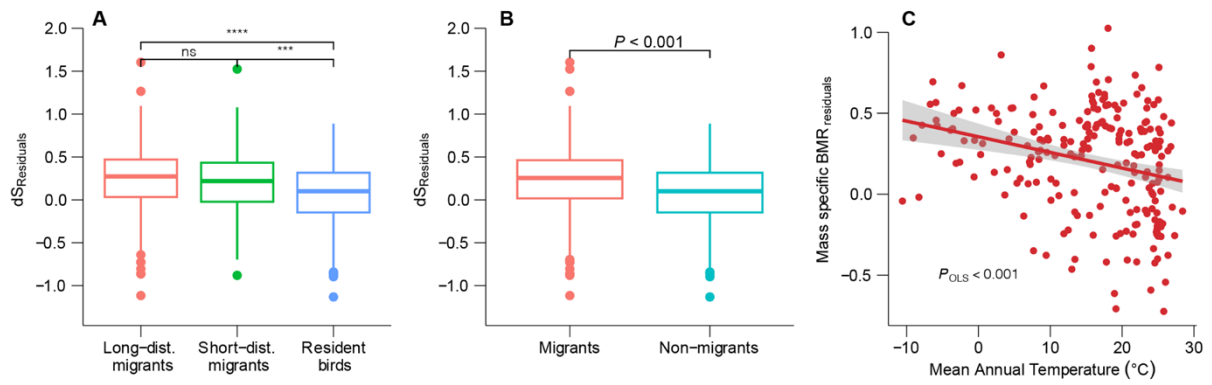


Fig. S16. Increased synonymous substitution rates (dS) of mitochondrial genes in migratory birds (A, B) and elevated mass-specific basal metabolic rates in birds from cold habitats (C). Average residuals of dS were estimated from the overall phylogenetic generalized least squares regression of $\log(dS)$ on $\log(\text{body mass})$. The residuals of the mass-specific basal metabolic rate (B) of birds were estimated by $\log B \sim \log M + 1/T_b$ to account for body mass (M), body temperature (T_b) and phylogenetic relatedness.

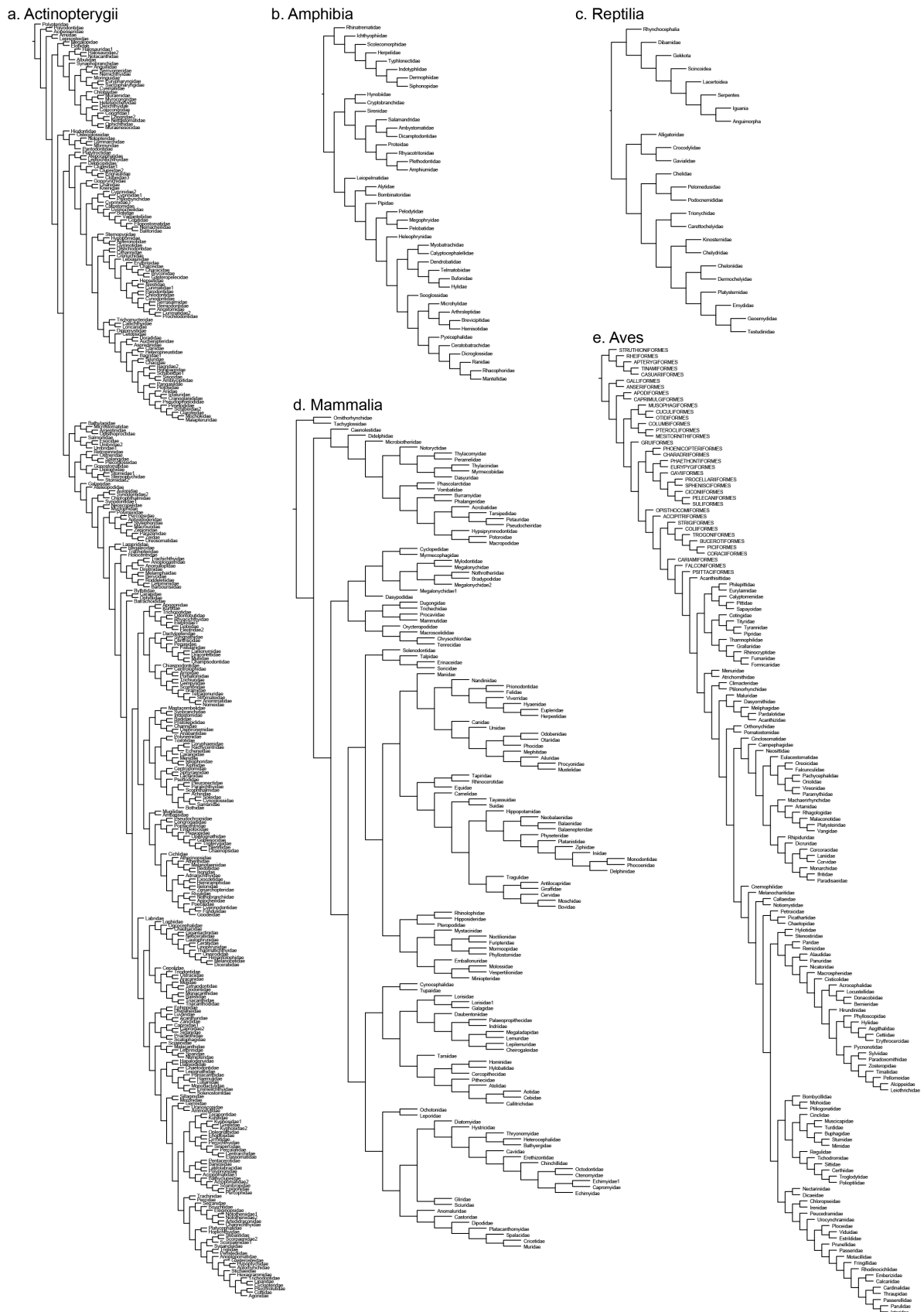


Fig. S17. Phylogenetic backbone trees were employed as topology constraints in the phylogenetic analysis using mitochondrial coding sequences. (A) Ray-finned fishes, (B) amphibians, (C) reptiles, (D) mammals, and (E) birds.

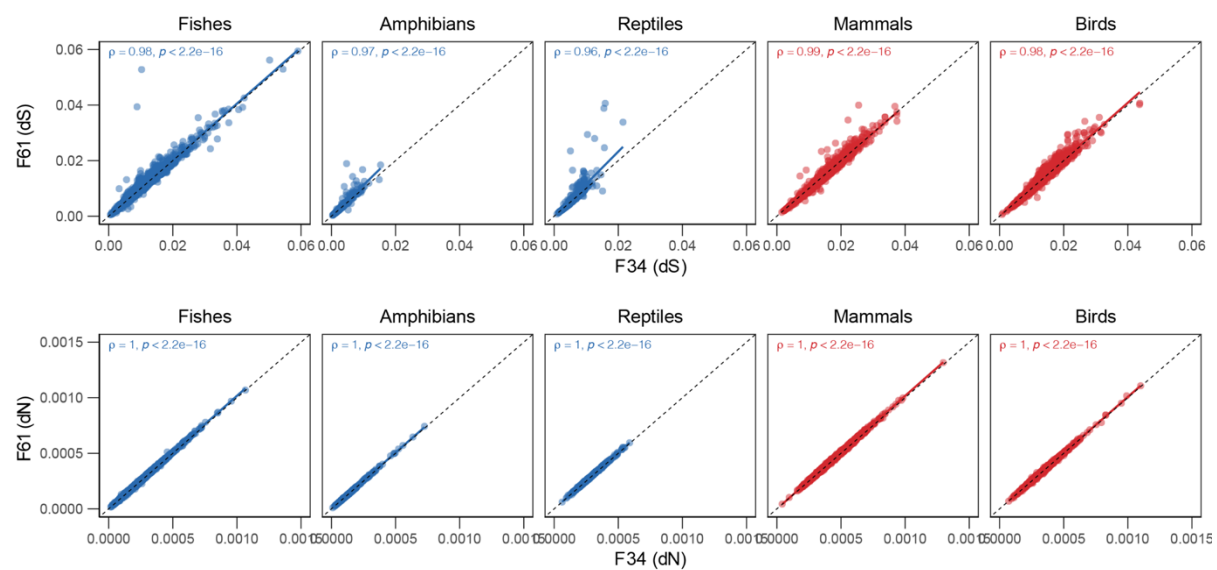


Fig. S18. Correlation plots illustrate the associations of molecular rates estimated by the F3x4 and F61 models. Spearman's rank correlation (ρ) was used to measure the correlation coefficients

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